

Evidence-Aware Validation of AI-Generated Drug Candidates Using PPI Networks and Network Pharmacology

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GitHub:

<https://github.com/Zander-Rothering/Evidence-Aware-Validation-of-AI-Generated-Drug-Candidates-Using-PPI-Networks>

Abstract

Generative AI models can produce structurally novel drug candidates, but they provide no mechanism to assess whether those compounds are biologically safe or clinically credible. We propose a post-generation validation pipeline that accepts an AI-generated SMILES string and an intended target protein (HMGCR), and delivers a structured HIGH, MEDIUM, or LOW risk classification by integrating compound-level similarity matching, PPI network-based off-target detection, and automated literature retrieval. The original goal was to test pre-existing AI-generated candidate libraries (ZINC250K), but the structural similarity of those compounds to known HMGCR inhibitors was poor (mean Tanimoto 0.225), so we fine-tuned MolGPT on 51 filtered ChEMBL HMGCR inhibitors as part of this work; this produced a substantial set of novel statin-like compounds as a useful by-product of the validation pipeline. The fine-tuned model produced 59 candidates (21.4% of novel SMILES) passing all four downstream filters, with the top hit reaching a Tanimoto of 0.58 against atorvastatin. A GNN trained on a STRING and BioGRID PPI subgraph centred on HMGCR produced a graph-level network risk score of 0.492. Together with an NLP-derived literature risk score, these signals are combined by an Evidence Aggregator into a final interpretable verdict.

1 Introduction

De novo drug discovery has historically been a slow and costly process, with drug candidates facing rigorous and stringent requirements to ensure safety and efficacy (Hinkson et al. 2020). The full development cycle averages 10 to 20 years and costs over \$1 to \$2 billion, with fewer than 10% of candidates achieving approval for clinical use (Sun et al. 2022). Many failures are detected only late into the process, costing companies and patients valuable time and resources (Das 2025). Therefore, there is a strong incentive for pharmaceutical companies to innovate drug development processes to improve the success rate of drug candidates, lower costs, and get medicines to the clinic more quickly (Hinkson et al. 2020). In response to these challenges, Artificial Intelligence has emerged as a promising strategy, with pharmaceutical companies increasingly integrating AI-driven approaches across the drug discovery pipeline to predict molecular properties, identify failures early, and ultimately improve the odds of clinical success (Niazi 2025).

Artificial intelligence has already begun to revolutionize the drug development process by focusing on the creation of novel compounds from scratch, reducing costs, and increasing the success rate of drug candidates from the lab to market (Tang et al. 2024). For instance, deep learning algorithms such as graph neural networks can quickly extract molecular structural features and identify relationships be-

tween these structures and toxicity, reducing months or years of data analysis to weeks (Zhang et al. 2025). Despite these advancements, AI-integrated drug development still faces notable challenges. While AI can accelerate discovery timelines and improve early safety predictions, challenges such as data quality, regulatory hurdles, model interpretability, and ethical concerns persist (Niazi 2025).

This project addresses a fundamental gap in AI-driven drug discovery: while generative models can produce structurally novel compounds, they provide no mechanism to assess whether those compounds are biologically safe or clinically credible. To bridge this gap, we propose a post-generation validation pipeline that accepts an AI-generated compound as a SMILES string alongside its intended target protein, and delivers a structured HIGH, MEDIUM, or LOW risk classification by integrating compound-level similarity matching, PPI network-based off-target detection, and automated literature retrieval.

1.1 Related Work

Generative models for de novo drug design have advanced rapidly, with transformer-based architectures such as MolGPT and reinforcement learning approaches such as REINVENT demonstrating the ability to produce novel SMILES strings with drug-like properties. MolGPT, introduced by Bagal et al. (2021), frames molecular generation as a condi-

tional language modelling problem, treating SMILES as character sequences and fine-tuning on target-specific compound sets to bias generation toward desired chemical space. While these models have shown strong performance on standard benchmarks such as MOSES and GuacaMol in terms of validity, uniqueness, and novelty, a critical gap remains: generated compounds are rarely subjected to systematic, evidence-grounded validation before being proposed as drug candidates. Existing post-generation filtering pipelines typically apply Lipinski Rule-of-Five and PAINS screens in isolation, without integrating structural similarity to known actives, literature evidence, or biological network context into a unified risk assessment. This project directly addresses that gap by building a multi-stream validation pipeline that produces an interpretable risk verdict backed by three independent evidence sources.

Network pharmacology has emerged as a principled framework for understanding how drugs interact with biological systems beyond their primary target. By representing proteins as nodes and their interactions as edges in a protein-protein interaction network sourced from databases such as STRING and BioGRID, researchers have shown that a compound’s therapeutic and adverse effect profile is shaped not only by its primary target but by the broader neighbourhood of interacting proteins. Graph Neural Networks have been applied to PPI networks to learn node embeddings that capture this neighbourhood context, enabling node-level classification tasks such as identifying off-target proteins associated with toxicity or unintended pathway activation (Jha et al. 2022). This project applies that paradigm specifically to HMGCR, building a PPI subgraph centred on the target and using a GNN to propagate risk signals from flagged neighbour proteins back to the query compound, producing a network-level risk score that complements the structural similarity and literature evidence streams.

Biomedical natural language processing has produced a class of domain-adapted transformer models, most notably BioBERT and PubMedBERT, that are pre-trained on large corpora of scientific literature and demonstrate strong performance on named entity recognition, relation extraction, and question answering tasks in the biomedical domain. Lee et al. (2020) showed that pre-training on PubMed abstracts and

full-text articles from PubMed Central produces representations that significantly outperform general-purpose BERT on biomedical benchmarks, making these models well-suited for extracting structured safety signals from unstructured clinical and pharmacological text. In the context of drug validation, literature evidence serves as a critical independent signal: a compound structurally similar to a known HMGCR inhibitor may still carry risk if published studies report adverse effects, off-target interactions, or toxicity concerns for its nearest-neighbour drug. This project incorporates an NLP Agent that queries PubMed using nearest-neighbor compound names, HMGCR target context, and SIDER-derived safety cues, applies a pretrained biomedical NER model to extract named entities and classify safety-relevant signals from retrieved abstracts, and converts those signals into a continuous literature risk score passed to the Evidence Aggregator alongside the structural similarity and network risk scores.

1.2 Pipeline Overview

The End-to-End Validation Pipeline takes an AI-generated SMILES string and routes it through three parallel evidence streams before producing a final risk verdict. Part 1, the Compound Matching Engine, parses the SMILES, encodes a 2048-bit ECFP4 fingerprint, and identifies the nearest-neighbour ChEMBL HMGCR inhibitor via Tanimoto similarity. It then applies drug-likeness filters, scaffold comparison, SIDER adverse-effect lookup, a rule-based structural tier, and a simple ANN RiskClassifier that provides a supporting ML tier and confidence flag. The MatchResult is passed simultaneously to Part 2A and Part 2B. Part 2A is an NLP Agent that queries PubMed using nearest-neighbor compound names, HMGCR target context, and SIDER-derived safety cues, extracts named entities from retrieved abstracts, and scores the literature evidence into a `literature_risk_score` between 0 and 1. Part 2B builds a PPI network from STRING and BioGRID, runs a GNN over the graph to learn node embeddings, and aggregates neighbour-level signals into a `network_risk_score` between 0 and 1. In Part 3, the Evidence Aggregator receives all three independent risk scores and combines them as a weighted sum, which the ValidationPipeline translates into a final HIGH, MEDIUM, or LOW verdict accompanied by a full evidence trail for interpretability.

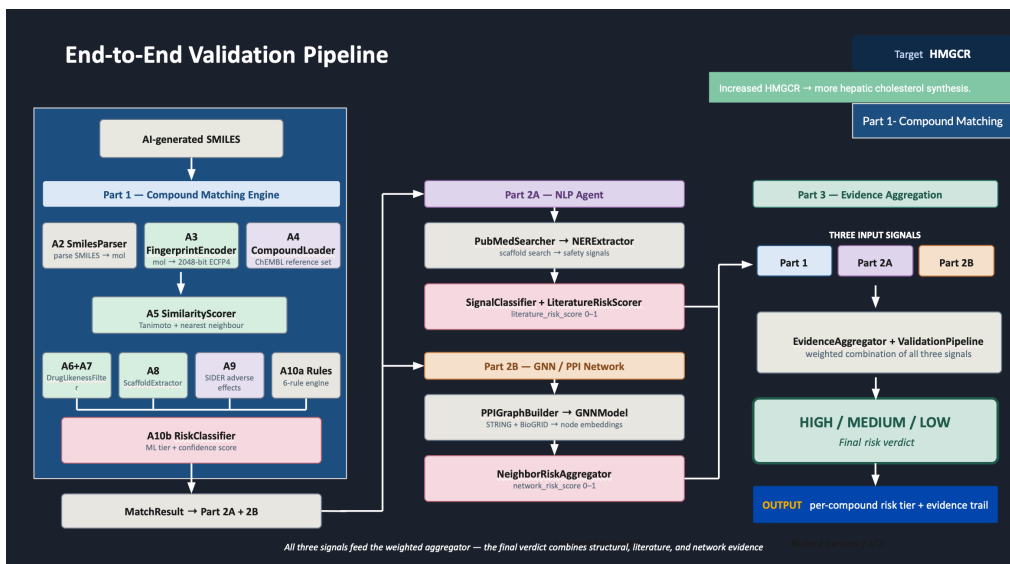


Figure 1: Overview of project pipeline from initial compound generation to final risk score generation.

2 Data

2.1 Compound Matching Engine

The Compound Matching Engine uses three data sources: ChEMBL402 HMGCR inhibitors as the reference library (up to 200 compounds via chembl-webresource-client), seven hardcoded marketed statins as an offline fallback (Atorvastatin, Simvastatin, Rosuvastatin, Fluvastatin, Pravastatin, Lovastatin, Pitavastatin), and MolGPT-generated candidate SMILES as queries. Each ChEMBL record provides a canonical SMILES, ChEMBL molecule identifier, and IC50 value retained as nearest-neighbour potency metadata. The query compounds are the output of MolGPT fine-tuned on filtered ChEMBL HMGCR inhibitors and represented as 2048-bit Morgan circular fingerprints at radius 2 generated via RDKit’s `rdFingerprintGenerator.GetMorganGenerator`; this 2048-bit vector serves both as the basis for Tanimoto nearest-neighbour search and as the primary feature input to the ANN classifier. The full data source summary is given in **Table 2**.

2.1.1 MolGPT Fine-Tuning for HMGCR Generation

Training data was sourced from ChEMBL target CHEMBL402 (HMG-CoA Reductase) and filtered through

four sequential layers: an activity filter retaining only IC50, Ki, and Kd measurements below 1 μ M; a pharmacophore filter requiring the HMG-mimetic warhead (open dihydroxy acid or δ -lactone) confirmed by SMARTS matching; a property window of MW 380 to 600, LogP 1 to 6, HBD 1 to 5, HBA 4 to 10, and rotatable bonds 6 to 14; and a PAINS screen. This yielded 51 filtered training compounds, expanded by RDKit-based SMILES enumeration augmentation with a strict molecule-level train/val split (zero parent overlap confirmed programmatically). The MolGPT v2 architecture used 4 transformer decoder layers, 4 attention heads, and embedding dimension 256 over a 78-token block size and 94-token chemical vocabulary. Training used AdamW (lr=1e-4, betas=(0.9, 0.95), weight decay 0.1), linear warmup over 10% of corpus tokens followed by cosine decay to 10% of peak, batch size 16, max 40 epochs with patience 5, initialised from a previously fine-tuned `statin_model.pt` checkpoint and run on a Mac M3 GPU using PyTorch’s MPS backend in fp32. Training loss converged from approximately 1.44 to a best validation loss of 0.5997 at epoch 37, with a final train-val gap of 0.081 indicating stable generalisation (**Fig. 2**); total wall time was approximately 55 minutes. Full configuration is summarised in **Table 5**.

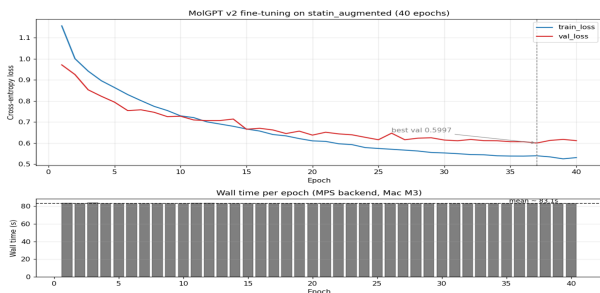


Figure 2: MolGPT v2 fine-tuning over 40 epochs; training loss converged from approximately 1.44 to a best validation loss of 0.5997 at epoch 37 with a final train-val gap of 0.081.

2.1.2 Generation and Filtering

5,000 candidate SMILES were sampled from the best checkpoint at temperature $T=1.0$ (batch size 128 on MPS). Candidates with disconnected fragments or fewer than 10 heavy atoms were removed at generation time. The validation pipeline applied four sequential filters: HMG warhead pharmacophore matching via SMARTS, Lipinski Rule-of-Five (one violation allowed), PAINS screening using RDKit’s canonical catalog, and Tanimoto similarity scoring against both the 51 filtered training compounds and 7 marketed statins using Morgan fingerprints (radius=2, 2048 bits). Of 5,000 generated SMILES, 276 were valid and novel after removing ex-

act ChEMBL reproductions; 59 (21.4%) passed all four filters (**Fig. 3**). The PAINS clean rate for novel candidates was 100%. The top hit achieved a Tanimoto of 0.58 against atorvastatin, representing a novel pyrrole-core statin with trifluoroaryl decoration and the canonical dihydroxy-acid tail. A two-panel summary in **Fig. 4** pairs the training loss curve with the candidate Tanimoto distribution split by warhead presence, showing warhead-positive candidates ($n=61$) concentrated in the right tail. A $T=0.8$ ablation on 2,000 samples produced higher mean Tanimoto (0.345 vs. 0.317) and warhead pass rate at the cost of reduced diversity; $T=1.0$ was retained for the final candidate set.

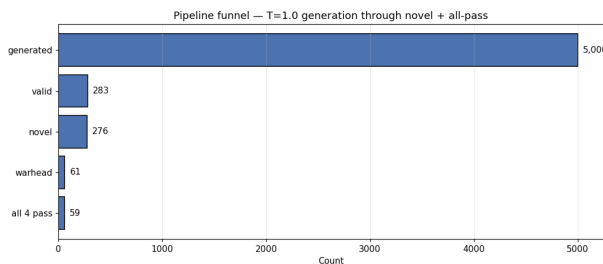


Figure 3: Generation funnel from 5,000 sampled SMILES at $T=1.0$ through validity (283), novelty (276), warhead pharmacophore (61), and the all-four-filter pass (59).

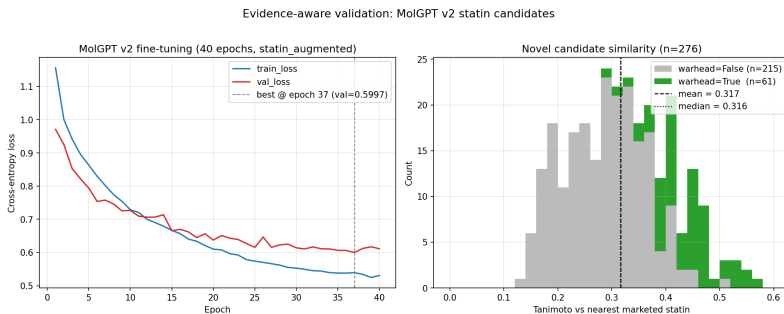


Figure 4: MolGPT v2 fine-tuning summary. Left: cross-entropy training and validation loss over 40 epochs, best checkpoint at epoch 37 (val=0.5997). Right: Tanimoto similarity distribution of the 276 novel candidates against the nearest marketed statin, split by warhead presence (warhead=False, $n=215$; warhead=True, $n=61$), with mean 0.317.

2.1.3 Feature Engineering

The MatchingEngine assembles per-compound features into the **ml_features** dictionary attached to each MatchResult (full enumeration in **Table 4**). The 2048-bit Morgan fingerprint serves as the basis for similarity computation; the Tanimoto coefficient against the nearest-neighbour ChEMBL reference and the scaffold-level Tanimoto on Murcko scaffolds capture global and core-ring similarity respectively. Seven physicochemical descriptors (MW, LogP, HBD, HBA, QED, TPSA, rotatable bonds) and three derived binary features (**lipinski_pass**, **pains_flag**, **brenk_flag**) provide bioavailability and structural-quality signal independent of fingerprint similarity. A binary **is_novel** feature, computed by the NoveltyChecker after canonicalising both query and training SMILES via RDKit, identifies exact reproductions of training compounds when paired with Tanimoto = 1.0.

2.2 NLP Agent

The NLP Agent is a literature-analysis orchestrator, not an autonomous LLM agent. It receives the MatchResult from the Compound Matching Engine, including the nearest-neighbor drug name, HMGCR target context, PubMed search terms, and SIDER-derived adverse-effect cues. PubMed abstracts are retrieved using the nearest-neighbor compound, HMGCR, and selected safety terms such as myopathy or hepatotoxicity. These abstracts are used for biomedical entity extraction, signal classification, and literature-based risk scoring.

2.3 PPI GNN

2.3.1 Edge Index and Edge Attributes

The PPI network used to train the GNN was constructed by integrating interaction data from the STRING database and BioGRID. Construction began by using a target protein (HMGCR) as a seed, from which interacting proteins were retrieved via STRING’s REST API. Interactions were filtered by taxonomic ID (Homo sapiens, 9606) and a confidence score threshold to retain high confidence associations. STRING interactions were merged with BioGRID interactions, which were processed separately under the same filtering conditions. Both sources were standardized to collect interacting protein pairs and interaction scores, while BioGRID additionally provided experimental system and experimental system type

data describing how interactions were detected. STRING interactions were assigned a default experimental system type of functional. After merging, duplicate interactions were removed and unique proteins were extracted. These proteins were assigned integer indices for node mapping and construction of the edge index tensor in the PyTorch Geometric Data object. Edge attributes were stored using normalized confidence scores along with label-encoded experimental system and experimental system type data.

2.3.2 Node Feature Vectors

The same set of unique proteins defining the PPI graph nodes was then used to retrieve biological feature information via the UniProt REST API. Extracted features included amino acid sequences and Gene Ontology (GO) IDs, representing biological process (BP), molecular function (MF), and cellular component (CC). The data were cleaned and parsed prior to embedding. Protein sequences were embedded using a pretrained Evolutionary Scale Modeling (ESM-2) language model, which generates dense embeddings by averaging token-level representations (Lin et al. 2023). GO terms were embedded using a Node2Vec model trained on a Gene Ontology graph, capturing structural relationships between GO identifiers (Grover and Leskovec 2016). Ontology specific embeddings (BP, MF, CC) were computed independently and aggregated. All features were then concatenated into a unified feature matrix and converted into a PyTorch tensor for input into the PPI GNN.

2.3.3 Node Target Labels

The same protein set was also used to construct node labels based on disease association from the DisGeNET database. A precompiled dataset of association scores was parsed and cleaned to produce a single normalized score (0 to 1) per protein. For proteins with multiple associations, the maximum score was retained to reflect the strongest evidence of disease association. Binary labels were then assigned by applying a threshold, where scores above the threshold were labeled as 1 (high disease association) and those below as 0 (low or no association). These labels were aligned with the node ordering and converted into a PyTorch tensor. Finally, all torch tensor edges, features, and labels were combined into a single PyTorch Geometric Data object for GNN training (**Fig. 8**).

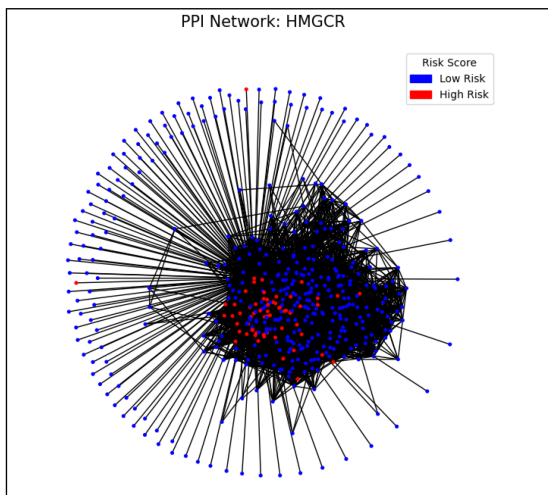


Figure 5: Graph representation of the HMGCN PPI network with nodes labeled by binary risk score (red = high risk, blue = low risk).

3 Methods and Models

3.1 Compound Matching Engine

The Compound Matching Engine transforms each generated SMILES into a `MatchResult` object for downstream validation. It parses and sanitizes the molecule with RDKit, generates a 2048-bit Morgan fingerprint, and identifies the nearest reference compound from the ChEMBL402-centered HMGCN library using Tanimoto similarity. It then applies drug-likeness filters, scaffold comparison, SIDER adverse-effect lookup, a rule-based structural tier, and an ANN RiskClassifier that provides a supporting ML tier and confidence flag. The engine is implemented as a sequence of composable classes (A1 through A10) shown in **Fig. 1**, each producing a typed intermediate output that the next stage consumes. Failures at any stage are logged and surfaced in the final evidence trail without halting the pipeline.

3.1.1 Input Parsing and Fingerprint Encoding

The **A2 SmilesParser** class wraps RDKit’s **MolFromSmiles** with strict validation: SMILES strings that fail to parse, contain disconnected fragments, or have fewer than 10 heavy atoms are rejected at this stage with a structured error rather than allowed to propagate as silent nulls. The parsed **Mol** object is then handed to the **A3 Finger-**

printEncoder, which generates the 2048-bit Morgan circular fingerprint at radius 2 using a single **rdFingerprintGenerator.GetMorganGenerator** instance shared across the pipeline. The encoder exposes both an RDKit **ExplicitBitVect** (used for Tanimoto computation) and a NumPy uint8 array (used as input to the sklearn classifiers) from the same generator call, eliminating the precision drift that would result from regenerating the fingerprint separately for similarity and ML feature extraction.

3.1.2 Reference Loading and Similarity Scoring

The **A4 CompoundLoader** initialises the ChEMBL reference set at pipeline startup by querying target ChEMBL402 through the chembl-webresource-client and caching the result in memory; the seven hardcoded marketed-statin SMILES serve as a deterministic offline fallback when the API is unreachable. The **A5 SimilarityScorer** consumes the query fingerprint and the cached reference set, computing pairwise Tanimoto coefficients via RDKit’s **BulkTanimotoSimilarity** and returning the nearest-neighbour ChEMBL compound along with the raw Tanimoto score. The Tanimoto coefficient between two binary fingerprints A and B is given by Equation 1, where a and b are the bit counts of A and B respectively and c is the count of shared bits. The bulk operation is vectorised at the C level by RDKit, allowing the full 200-compound reference set to be scored in under 10 ms per query.

$$T(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{c}{a + b - c} \quad (1)$$

3.1.3 Drug-Likeness, Scaffold, and Adverse-Effect Filters

Three orthogonal filters operate in parallel on the parsed molecule. The **A6 DrugLikenessFilter** computes the seven physicochemical descriptors (MW, LogP, HBD, HBA, QED,

TPSA, rotatable bonds) using RDKit’s **Descriptors** and **rdMolDescriptors** modules, then applies the four-rule Lipinski check and the PAINS and Brenk structural-alert catalogs via RDKit’s **FilterCatalog**. A7 binarises the resulting flags into the **lipinski_pass**, **pains_flag**, and **brenk_flag** fields.

The **A8 ScaffoldExtractor** computes the Murcko scaffold of both the query and the nearest neighbour using **MurckoScaffold.GetScaffoldForMol**, encodes each scaffold as a separate Morgan fingerprint, and returns a scaffold-level Tanimoto coefficient that captures core ring-system overlap independent of sidechain decoration. The **A9 SIDER lookup** maps the nearest-neighbour ChEMBL ID to a precompiled SIDER (Side Effect Resource) table and returns the count of reported adverse effects, which serves as a literature-grounded prior on the safety profile of the structurally closest drug.

3.1.4 Rule Engine and Risk Classification

The **A10a Rules** module implements a six-rule decision engine that produces a deterministic risk tier as an interpretable counterweight to the ML classifier. The rules are evaluated in order: (1) Tanimoto ≥ 0.95 with **is_novel = False** flags an exact reproduction of a training compound; (2) Tanimoto < 0.30 flags structural divergence from the known active class; (3) two or more Lipinski violations flag oral bioavailability concerns; (4) a positive PAINS or Brenk flag flags reactive scaffold liability; (5) a SIDER adverse-effect count above the cohort median elevates the risk tier; and (6) QED < 0.34 applies a HIGH-risk bump for poor composite drug-likeness. Each rule writes its verdict to the evidence trail with the triggering value, so the final tier can be audited stage-by-stage.

The **A10b RiskClassifier** uses similarity-derived features to provide a supporting ML tier and probability. It is a simple PyTorch ANN trained on weak rule-derived structural labels. This model uses the engineered feature set summarized in **Table 4**, including Tanimoto similarity, scaffold similarity, drug-likeness descriptors, structural alert flags, and potency-related metadata. Training labels are weak rule-derived structural labels from Part 1 features, while IC50 contributes to potency-related feature engineering. The model is evaluated with a stratified 20% split on the 495 prescreened EDA molecules. At inference, the ANN returns a supporting ML tier and model probability, which are compared with the rule-based tier to produce the final Part 1 confidence flag.

3.1.5 MatchResult Output Contract

The final **MatchResult** dataclass carries six fields downstream to the NLP Agent and PPI GNN: the canonical query SMILES, the nearest-neighbour ChEMBL ID and Tanimoto score, the full **ml_features** dictionary, the rule-engine tier,

the classifier risk score with confidence, and the structured evidence trail. This contract is enforced via Python type hints and validated at construction time; downstream stages can therefore consume the object without re-parsing or re-validating the underlying SMILES.

3.2 NLP Agent

The NLP Agent converts each MatchResult into a literature risk score using the nearest-neighbor compound, HMGCR target context, and SIDER-derived safety cues. Before PubMed lookup, the pipeline tries to convert ChEMBL nearest-neighbor IDs into readable drug names. Exact SMILES matches to the seven named statins are replaced directly with the statin name. Otherwise, unresolved ChEMBL IDs are mapped to the closest marketed statin only when the generated molecule has Tanimoto similarity of at least 0.40. This prevents weakly related novel molecules from inheriting overly specific literature evidence from an approved statin. After the lookup name is chosen, SIDER files provide adverse-effect cues, and PubMed abstracts are retrieved through NCBI Entrez using fallback queries that broaden from compound-specific HMGCR searches to statin-class or HMGCR inhibitor safety literature when needed. Retrieved abstracts are processed with the pretrained biomedical NER model `d4data/biomedical-ner-all`, mapped into efficacy or safety signals, and converted by the LiteratureRiskScorer into a 0–1 literature risk score for the Evidence Aggregator.

3.3 GNN Model

The GNN model utilized for our PPI predictive modeling was implemented using the PyTorch Geometric message passing framework, which enables automated propagation of information across graph-structured data. The model consists of message passing layers, each performing a linear transformation of node feature vectors followed by message aggregation of neighboring nodes. Each layer applies a learned linear transformation to node features, after which messages are propagated along edges using degree normalization to prevent highly connected nodes from dominating the learning process. Self-loops are added to incorporate each node’s own features during propagation, and a learnable bias term is applied to the aggregated messages. Message aggregation is performed via summation. The forward propagation rule for a single GNN layer is given in Equation 2.

$$\mathbf{x}_i^{(k)} = \sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{\deg(i) \cdot \deg(j)}} \cdot \left(\mathbf{W}^\top \cdot \mathbf{x}_j^{(k-1)} \right) + \mathbf{b} \quad (2)$$

The base GNN architecture consists of three message passing layers, each followed by a ReLU activation function and dropout layer. This design enables information to be propagated across each node’s immediate neighbor as well as incorporate information about its neighbors-neighbor. The resulting node embeddings are passed to a linear classifier that

outputs logits for node level classification.

Model training was conducted using a randomized node level split of 50% training, 30% validation, and 20% test data. Optimization was performed via the Adam optimizer with cross-entropy loss, using fixed learning rate and weight decay parameters. During training, gradients were computed via

backpropagation and model performance was monitored on the validation set, with the best-performing model retained. Final node level predictions were obtained by applying a softmax function to the output logits to produce class probabilities, which were interpreted as risk scores. A graph level risk score was then computed as the mean of all node level risk scores.

3.4 Risk Aggregation

The risk aggregator serves as the final decision layer of the pipeline, integrating the three evidence streams into a unified assessment. Each input risk score is weighted and combined into a single normalized score using a weighted sum. This combined score is assigned to a final risk tier, High/Medium/Low, based on the agreement across the individual evidence streams. The final output preserves the aggregated score along with all intermediate evidence to support analysis of the end-to-end pipeline.

4 Results and Evaluation

The final ValidationPipeline integrates evidence from the Compound Matching Engine, the NLP Agent literature modules, and the PPI/GNN network modules. For each generated SMILES, the Compound Matching Engine provides the nearest HMGCR reference similarity and drug-likeness signals, the NLP Agent provides a literature risk score, and the PPI/GNN module contributes a target-level network risk score. The three evidence streams are combined using fixed weights defined in the RiskWeightConfig class to produce a final LOW, MEDIUM, or HIGH risk tier for each candidate.

4.1 Compound Matching Engine Results

The Compound Matching Engine produced the primary structural evidence stream for all 495 evaluated MolGPT candidates. It assigned nearest-neighbor similarity scores, drug-likeness filter outputs, scaffold overlap features, SIDER safety cues, and an initial structural risk tier for downstream aggregation. In the final LOW-risk set, all 17 candidates passed Lipinski and PAINS filters, and 10 retained the expected HMGCR pharmacophore. The LOW-risk set overlapped more strongly with the ChEMBL-based top16 structural screen than with the marketed-statin-only top16 screen, indicating that the structural stream favored broader HMGCR-like chemical space rather than simply selecting candidates most similar to approved statins.

4.2 NLP Agent Results

The NLP Agent generated literature risk scores for all 495 pre-filtered MolGPT generated candidates. For 173 candidates, PubMed retrieval used the most specific search strategy based on the nearest-neighbor compound and HMGCR target context. For the remaining 322 candidates, the NLP pipeline used a broader statin-class literature search because compound-specific proxy evidence was not available or not appropriate. Literature risk scores ranged from 0.1500 to 0.8675, with a mean score of 0.6121. Every candidate thus received a literature risk score, with the pipeline recording whether each score was derived from a specific nearest-neighbor PubMed search or from broader statin-class fallback evidence.

4.3 PPI GNN Results

The PPI network constructed from STRING and BioGRID consisted of 479 nodes and 9482 edges, representing proteins in the HMGCR interaction neighborhood and the experimentally validated or computationally predicted interactions between them. As previously mentioned, node feature vectors were constructed by encoding UniProt Gene Ontology annotations as binary vectors, and binary node labels were assigned using DisGeNET disease association scores with a threshold of 0.3. The dataset was split into 50% training, 30% validation, and 20% test nodes for model evaluation. The resulting graph structure is visualized in **Fig. 8**.

The GNN model was trained for 1,000 epochs using the Adam optimizer with cross-entropy loss. Training loss decreased from 0.6842 to 0.4385 and validation loss decreased from 0.6813 to 0.4353, with both curves converging closely throughout training and showing no evidence of overfitting (**Fig. 9**). Validation accuracy reached 88.91% at epoch 1 and remained constant across all 1,000 epochs (**Fig. 10**), indicating that the model learned to predict the majority class from the beginning. This behavior is consistent with the class imbalance present in the node labels, where proteins labeled low risk significantly outnumber those labeled high risk, causing the model to favor majority class predictions. Despite this limitation, the steadily decreasing loss confirms that the model extracted meaningful signals from the graph structure and node features. The final graph-level network risk score for the HMGCR PPI network was 0.492, reflecting a moderate biological risk signal based on the aggregated node-level predictions across the interaction neighborhood.

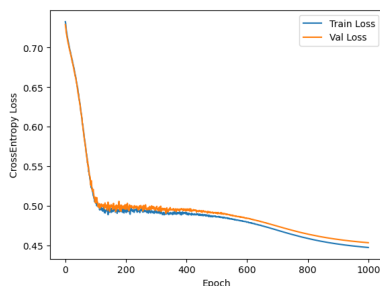


Figure 6: Training and validation cross-entropy loss during GNN training on the HMGCR PPI network over 1,000 epochs. Both curves converge closely without evidence of overfitting.

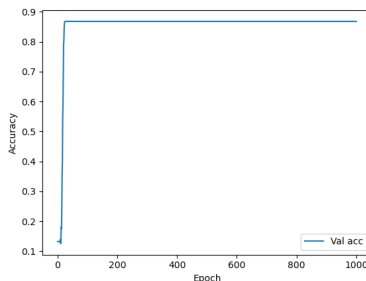


Figure 7: Validation accuracy of the GNN node classification model over 1,000 training epochs on the HMGCR PPI network. Accuracy plateaus at 88.91% from epoch 1, reflecting class imbalance in node labels.

4.4 Risk Aggregation Results

The final risk aggregation of the end-to-end pipeline identified 17 LOW-risk candidates from the full validation set of 495 molecules. Of these, 16 passed the PubChem novelty check, while one was flagged as reference-like/non-novel. All 17 passed Lipinski and PAINS filters, and 10 also retained the expected HMGCR pharmacophore, making this subset the strongest rule-consistent candidate group.

Compared with the initial structural screens, the LOW-risk set overlapped strongly with the ChEMBL-based top 16 list but only weakly with the marketed-statin top16 list, showing that the final pipeline was not simply copying the approved-statin similarity ranking. From these 17 candidates, we selected a stricter Top 5 shortlist based on novelty, pharmacophore retention, filter passing, high-confidence agreement, stronger ChEMBL similarity, and lower aggregate risk score.

The final Top 5 shortlist was primarily supported by the ChEMBL-based structural screen, with two candidates also appearing in the marketed-statin top16 list: candidate 1 ranked 1st by ChEMBL similarity and 15th by marketed-statin similarity, while candidate 5 ranked 13th by ChEMBL similarity and 9th by marketed-statin similarity. This overlap provides an additional sanity check that the final shortlist preserves approved-statin-like relevance without simply reproducing the marketed-statin similarity ranking. The final five SMILES are listed in **Table 1**.

5 Discussion

5.1 Compound Matching Discussion

A key limitation of the Compound Matching Engine is that structural similarity is only a proxy for biological validity. The final candidates were filtered to remove exact ChEMBL duplicates and were additionally annotated with PubChem novelty, but the structural reference library was still centered on ChEMBL402 HMGCR inhibitors. Therefore, the matching score should be interpreted as HMGCR-like structural support rather than fully independent experimental validation. This was an appropriate primary reference choice because ChEMBL402 provides a broad target-specific inhibitor library, whereas using only marketed statins would over-bias the ranking toward approved-drug similarity. However, some nearest neighbors remained less clinically interpretable ChEMBL IDs, and an independent activity source such as BindingDB would be needed to further calibrate the similarity threshold against experimental HMGCR activity.

5.2 NLP Discussion

The main limitation of the NLP pipeline is that its literature score is proxy evidence rather than direct evidence for each generated molecule. Because novel SMILES typically do not appear in PubMed, the pipeline searches using the nearest-neighbor compound name, HMGCR context, and SIDER-derived safety cues instead of the query SMILES itself. This

proxy-based design reduces misleading drug-name transfer for distant analogs, but it also means that some NLP scores reflect class-level HMGCRC evidence rather than molecule-specific literature. This design gives novel compounds some benefit of the doubt while still capturing known statin safety themes.

5.3 PPI GNN Discussion

The GNN-derived risk score of 0.492 for the PPI network indicates a moderate biological risk signal, reflecting the aggregation of node-level predictions across the interaction network. Model performance shows a rapid increase in accuracy during the initial training epochs, followed by a plateau. This behavior suggests that the model quickly learns dominant patterns in the data, particularly those associated with the majority class. The continued decrease in loss despite stable accuracy indicates improved confidence in predictions rather than improved predictive ability, as the model improves the estimated probability of each node without changing predicted labels. As noted in the results, this pattern is partially driven by class imbalance, which is likely inflating accuracy by favoring the predominant low-risk class.

This class imbalance is one of several limitations of the current GNN framework that may hinder its predictive performance. The class imbalance creates a bias in the model toward low-risk predictions, reducing its predictive ability for higher-risk proteins. This issue is compounded by the use of an arbitrary threshold for label assignment, which may not align with biologically validated risk definitions.

Additionally, the model does not incorporate edge attributes. Currently, our graph is undirected, propagating messages between nodes indiscriminate of the experimental system type and interaction score. By utilizing these edge attributes and selecting features with higher predictive capabilities, further optimization could be achieved. Finally, while the overall risk score for our PPI network is currently calculated as an aggregate of node-level scores, the model is

trained on node-level classification. Consideration is required as to whether we should transition to training at the graph level, with loss calculated after a final risk score aggregation is complete, to better align the training objective with the final output.

5.4 Conclusion

The final pipeline identified 17 LOW-risk candidates from the full validation set of 495 molecules. Of these, 16 passed the PubChem novelty check, while one was flagged as reference-like/non-novel. All 17 passed Lipinski and PAINS filters, and 10 also retained the expected HMGCRC pharmacophore, making this subset the strongest rule-consistent candidate group. Compared with the initial structural screens, the LOW-risk set overlapped strongly with the ChEMBL-based top16 list but only weakly with the marketed-statin top16 list, showing that the final pipeline was not simply copying the approved-statin similarity ranking. From these 17 candidates, we selected a stricter Top 5 shortlist (**Table 1**) based on novelty, pharmacophore retention, filter passing, high-confidence agreement, stronger ChEMBL similarity, and lower overall risk.

The strong overlap between the final LOW risk candidates and the ChEMBL similarity rankings suggests that the pipeline successfully identified compounds with biologically relevant statin-like features, while the weaker overlap with marketed statins indicates that the workflow was not simply reproducing approved-drug similarity. The consistent passing of Lipinski and PAINS filters, along with retention of the HMGCRC pharmacophore in most candidates, further supports the reliability of the validation pipeline. Overall, combining structural similarity, pharmacophore validation, novelty assessment, and aggregated risk scoring produced a more selective and biologically plausible candidate set than structural screening alone.

Table 1: Final Top 5 LOW-risk candidate SMILES from the end-to-end ValidationPipeline.

#	SMILES
1	<chem>CC(C)c1nc(-c2ccccc2)c(-c2ccc(F)cc2)n1C=CC(=O)CC(=O)CC(=O)O</chem>
2	<chem>CC(C)c1nc(-c2ccc(F)cc2)c(-c2ccccc(F)c2)n1C=CC(=O)CC(=O)CC(=O)O</chem>
3	<chem>CC(C)c1nc(-c2ccccc(F)c2)c(-c2ccc(F)cc2)n1C=CC(=O)CC(=O)CC(=O)O</chem>
4	<chem>CC(C)c1c(-c2ccc(F)cc2)nc(-c2ccccc(F)c2)n1C=CC(=O)CC(=O)CC(=O)O</chem>
5	<chem>CC(C)c1nc(-c2ccc(F)cc2)n(C=CC(=O)CC(=O)CC(=O)O)c1C(C)C</chem>

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Appendix

A. MolGPT Training Challenges

MolGPT fine-tuning encountered fifteen technical challenges spanning installation, software compatibility, data formatting, model checkpointing, and hardware portability. Installation was complicated by a pomegranate C++ compile failure under Python 3.13, an unrelated PyPI package occupying the `moses` namespace, and missing `git-lfs` preventing dataset checkout. Software compatibility issues required patching the MOSES codebase for the `rdkit.six` removal in newer RDKit releases and the `DataFrame.append` removal in pandas 2.0, as well as fixing a `learning_rate` type=int bug in `train.py` that rejected float values. Training data preparation required regenerating the input CSV with the four mandatory columns (`smiles`, `qed`, `scaffold_smiles`, `source`) using the value `val` rather than `test` for the validation split, stripping stereochemistry tokens absent from the 94-token vocabulary, and filtering disconnected SMILES. Post-training, a positional-embedding versus attention-mask shape mismatch (151 vs. 253) blocked checkpoint loading until the mask keys were removed and `strict=False` was passed to `load_state_dict`. Finally, the generation script's hardcoded CUDA calls were replaced with `.to('mps')` to run on the Apple M3 GPU, and the missing `statin_stoi.json` vocabulary file was reconstructed from the hardcoded `whole_string` list in `train.py`. A separate runtime constraint emerged when an initial 200-epoch run at batch size 32 was force-killed by macOS at epoch 134 after consuming 96.82 GB of system memory, which was traced to PyTorch MPS tensor accumulation across epochs and resolved by reducing to 40 epochs at batch size 16 with explicit `torch.mps.empty_cache()` calls between epochs.

B. GNN Training Challenges

The main challenges encountered during training of our PPI GNN involved parsing and integrating biological datasets into a unified network representation. The datasets extracted from STRING, BioGRID, and UniProt did not share a consistent format suitable for direct use in the GNN. Extensive preprocessing was required to standardize protein identifiers and construct the node and edge representations used for graph learning. Another challenge arose when generating meaningful node features from UniProt data. Initial attempts to use label encoding for categorical annotations failed to capture biological relationships that could be effectively learned by the GNN. To improve feature quality, embedding-based approaches were adopted instead. Protein sequences were embedded using the ESM-2 protein language model to capture sequence-level relationships, while Gene Ontology (GO) identifiers were embedded using Node2Vec to preserve relationships between GO terms. Finally, the initial DisGeNET label assignment using a 0.5 threshold created an extreme class imbalance with only 12 of 495 nodes labeled high-risk, causing the GNN to collapse to majority-class predictions. Lowering the threshold to 0.3 produced the class distribution used for the reported experiments, although class imbalance remained a key limitation as discussed in §6.3.

C. Compound Matching Engine Data Sources

Table 2: Data sources used by the Compound Matching Engine.

Data Source	Records	Role	Used By
ChEMBL CHEMBL402	Up to 200	Reference / nearest-neighbor metadata	CompoundLoader, SimilarityScorer
7 named statins (fallback)	7	Offline reference baseline	CompoundLoader
MolGPT Run 2 candidates	99 clean	Query compounds	MatchingEngine input
MolGPT training set	665 SMILES	Novelty check baseline	NoveltyChecker

D. Drug-likeness descriptor ranges

Table 3: Drug-likeness descriptor ranges across the candidate hit set, with Lipinski risk thresholds.

Descriptor	Lipinski Limit	Mean (hits)	Min	Max	Risk Rule
MW (Da)	≤ 500	~ 420	280	560	$> 500 =$ violation
LogP	≤ 5	~ 3.8	1.2	6.1	$> 5 =$ violation
HBD	≤ 5	~ 2.1	0	5	$> 5 =$ violation
HBA	≤ 10	~ 5.4	2	9	$> 10 =$ violation
QED	> 0.34	~ 0.62	0.28	0.87	$< 0.34 =$ HIGH bump
TPSA ($\text{\AA}^2$)	< 140	~ 98	55	138	Informational

E. Matching Engine Feature Set

Table 4: Feature set assembled by the MatchingEngine for each query compound.

Feature	Type	Source Class	ML Role
Morgan fingerprint (2048-bit)	Binary vector	FingerprintEncoder	Similarity feature input
Tanimoto (vs NN reference)	Float [0,1]	SimilarityScorer	Risk threshold + feature
Scaffold Tanimoto	Float [0,1]	ScaffoldExtractor	Core similarity feature
MW, LogP, HBD, HBA, QED, TPSA, rot_bonds	Float / Int	DrugLikenessFilter	Bioavailability features
lipinski_pass	Binary	DrugLikenessFilter	Hard filter signal
pains_flag, brenk_flag	Binary	DrugLikenessFilter	Hard filter signal
is_novel	Binary	NoveltyChecker	Memorisation detector

F. MolGPT Fine-Tuning Configuration

Table 5: MolGPT v2 fine-tuning configuration (final run).

Field	Value
Architecture	MolGPT (cond_gpt branch), 4 layers, 4 heads, 256 embedding dimension
Vocab / block	94 tokens / 78-token block
Optimizer	AdamW, lr=1e-4, betas=(0.9, 0.95), weight decay 0.1
LR schedule	Linear warmup (10% of corpus tokens) $\rightarrow$ cosine decay to 10%
Batch size	16
Max epochs	40 (early-stop patience 5)
Device	Apple MPS (M3, fp32; autocast/GradScaler unstable on MPS)
Init from	cond_gpt/weights/statin_model.pt (ChEMBL402 fine-tuned, 13.4 MB)

G. Pipeline Class Architecture Table

Class	Name	Role
A1	ValidationPipeline	End-to-end orchestrator; routes input SMILES through Parts 1, 2A, 2B and aggregates results
A2	SmilesParser	Parses SMILES via RDKit MolFromSmiles; rejects fragments and small molecules
A3	FingerprintEncoder	Generates 2048-bit Morgan radius-2 fingerprint; emits ExplicitBitVect and uint8 array
A4	CompoundLoader	Loads ChEMBL ChEMBL402 reference set via API; falls back to 7 marketed statins offline
A5	SimilarityScorer	Computes Tanimoto via BulkTanimotoSimilarity; returns nearest-neighbour reference
A6 / A7	DrugLikenessFilter	Computes MW, LogP, HBD, HBA, QED, TPSA, rot bonds; applies Lipinski, PAINS, Brenk
A8	ScaffoldExtractor	Extracts Murcko scaffold; computes scaffold-level Tanimoto
A9	SIDER lookup	Maps nearest-neighbour ChEMBL ID to SIDER adverse-effect counts
A10a	Rules engine	Six-rule deterministic risk tier based on Tanimoto, Lipinski, PAINS/Brenk, SIDER, QED
A10b	RiskClassifier	Simple ANN RiskClassifier; outputs similarity-informed ML tier and model probability
B1	PubMedSearcher	Queries PubMed using nearest-neighbor names, HMGCR, and SIDER cues
B2	NERExtractor	Pretrained biomedical NER extracts entities from retrieved abstracts
B3	SignalClassifier	Classifies safety-relevant signals from extracted entities
B4	LiteratureRiskScorer	Aggregates signals into literature_risk_score 0–1
C1	PPIGraphBuilder	Builds STRING + BioGRID subgraph centred on HMGCR
C2	GNNModel	Three-layer message-passing GNN; node-level binary classification
C3	NeighborRiskAggregator	Aggregates node-level predictions into network_risk_score 0–1
D1	EvidenceAggregator	Weighted combination of three risk scores
D2	MatchResult / RiskTier	Output dataclass: per-compound HIGH/MEDIUM/LOW verdict + evidence trail

H. Top 16 novel candidates

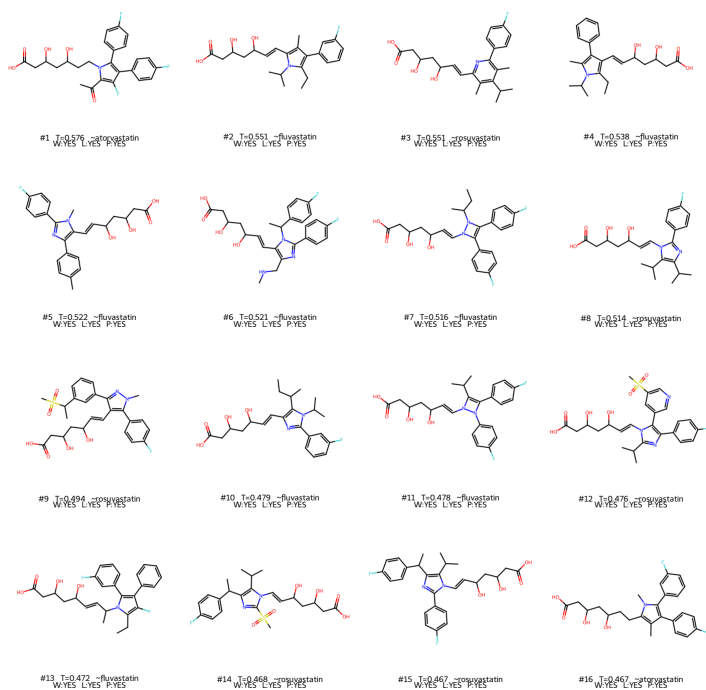


Figure 8: Top 16 novel candidates ranked by Tanimoto similarity to the nearest marketed statin, with warhead (W), Lipinski (L), and PAINS (P) flags shown for each. The top hit (#1, T=0.576 vs. atorvastatin) features a pyrrole core with trifluoroaryl decoration and the canonical dihydroxy-acid tail. Most subsequent hits cluster around fluvastatin and rosuvastatin scaffolds; all 16 candidates pass the warhead, Lipinski, and PAINS filters.